

SafeRL-Drive: Traffic-Aware Reinforcement Learning in Procedural Driving Scenarios

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Abstract

SafeRL-Drive studies whether a continuous-control policy that has learned procedural road geometry can adapt to moving traffic without forgetting traffic-free driving. The learned Soft Actor-Critic (SAC) policy controls one ego vehicle; MetaDrive’s native traffic manager controls all surrounding vehicles. Earlier controls established a reproducible pipeline and showed that a staged geometry curriculum raises held-out success from 12% to 87% relative to equal-budget direct SAC. The same frozen curriculum policies achieved 69% success but 21% collision under a zero-shot density-0.05 traffic shift, motivating one bounded extension. Two seed-0 warm-start pilots differ only in collision penalty; a predeclared rule selects one for a seed-1 confirmation. Final evaluation pairs both lineages on 100 fixed scenarios at densities 0, 0.05, and 0.10 and reports success, collision, route completion, and traffic-free retention. This document distinguishes the verified geometry evidence from traffic results that still require the planned Colab runs; it makes no real-world safety claim.

1 Introduction

Closed-loop simulation exposes a driving policy to the consequences of its own actions. MetaDrive combines procedural road generation, reactive background traffic, physics, and vector observations in a lightweight environment suited to generalization experiments [2]. SafeRL-Drive uses these features for a narrow student research question:

Can a geometry-competent SAC policy adapt to interactive procedural traffic while reducing collisions and retaining traffic-free road-generalization ability?

The claim is intentionally smaller than autonomous driving. The project does not model real sensors, perception uncertainty, pedestrians, adverse weather, or deployment risk. It tests reproducible learning and retention in one simulator.

2 Single-Agent Traffic Formulation

The task is single-agent reinforcement learning. SAC controls one ego vehicle with continuous steering and throttle/brake. MetaDrive independently spawns and controls background traffic with its native policies. Every condition fixes `num_agents=1` and `is_multi_agent=false`; no surrounding policy is trained.

The policy receives MetaDrive’s default bounded LidarState vector: ego state, navigation and road fea-

tures, and lidar information about nearby vehicles. Training never uses an RGB observation or CNN. A separate recording environment activates the moving main camera while retaining the same vector observation for policy inference.

All tasks use three-block procedural maps, a 1,000-step horizon, fixed spawn lanes, continuous full-range actions, and zero accident probability. Training cycles through 200 scenarios from seed 40000. Validation starts at 50000; the held-out test starts at 60000. Evaluation traffic is deterministic conditional on its scenario seed (`random_traffic=false`, `traffic_mode=respawn`).

3 Geometry Curriculum as Initialization

Phase 1 established the infrastructure on an intentionally easy straight, traffic-free road. IDM, PPO, and SAC each reached 100% success on 20 held-out episodes. Because PPO used a discrete action grid while SAC used continuous control, these were pipeline controls rather than a fair algorithm ranking.

Phase 2 asked whether staged road geometry helps SAC generalize. Direct SAC trained on three-block procedural maps. Curriculum SAC trained first on curves, then straight-plus-curve roads, and finally on the same procedural target. Architecture, reward, action space, maximum 500,000-step budget, and test split matched.

The curriculum improved success by 75 percentage

Table 1: Verified Phase-2 held-out means across two completed seeds.

Condition	Succ.	Coll.	Off-road	Route
Direct SAC	12.0%	15.5%	83.0%	47.0%
Curriculum SAC	87.0%	0.0%	9.5%	95.1%

Table 2: Controlled seed-0 traffic pilots.

Setting	Reference	Safety
Vehicle collision penalty	5	10
Object collision penalty	5	10
Success reward	10	10
Off-road penalty	5	5
New transitions	300k	300k

points and route completion by 48 points (Table 1). Seed 0 improved from 24% to 86% success and seed 1 from 0% to 88%. A third curriculum launch failed its mandatory curve gate and is retained as negative evidence. These results motivate warm start, not a claim that the optimizer is uniformly reliable.

The frozen policies were then evaluated without traffic training at density 0.05. Direct SAC averaged 3.5% success and 37% collision. Curriculum SAC averaged 69% success, 21% collision, 6% off-road, and 87.1% route completion. This is promising transfer but not traffic competence.

4 Traffic-Adaptation Design

The extension uses exactly three long runs. Both seed-0 pilots start from the corresponding best geometry-curriculum checkpoint and receive fresh traffic replay buffers. Their only intended learning difference is collision penalty:

Each pilot uses 100,000 maximum transitions at density 0.02 followed by 200,000 at density 0.05. Stage A must reach 70% success, 80% route completion, at most 25% collision, and at most 15% off-road on 25 fixed validation scenarios. A miss pauses the lineage; it does not trigger another pilot.

SAC [1] is implemented with Stable-Baselines3 [3] and uses an MLP with two 256-unit layers, learning rate 10^{-4} , a 300,000-transition buffer, 5,000 learning-start transitions, batch size 512, discount 0.99, target smoothing 0.005, fixed entropy coefficient 0.05, and one-step update frequency. Parallel simulator processes are capped at four, and gradient steps equal the number of environments so the update-to-data ratio remains approximately stable.

At the end of seed 0, both pilots are evaluated at densities 0, 0.05, and 0.10 on 50 validation scenarios. Qualification at density 0.05 requires at least 80% success and 90% route completion, at most 10% collision and off-road, and no more than a ten-point traffic-free success loss. If both qualify, lower collision wins, with success and then route completion as declared tie-breakers. If neither qualifies, higher primary success wins unless within five points, when lower collision wins. Only the selected condition is confirmed from the seed-1 geometry checkpoint.

5 Evaluation Protocol

Frozen source and selected adapted policies from seeds 0 and 1 are evaluated on identical scenario IDs at densities 0, 0.05, and 0.10. IDMPolicy and ExpertPolicy run on the same matrix. Each policy-density condition contains 100 deterministic episodes.

Per-episode output retains MetaDrive’s overlapping raw flags and adds one exclusive terminal outcome: success, collision, off-road, timeout, or other. Summaries include a Wilson 95% interval for success; aggregate and vehicle-specific collision; collision-free, off-road, and timeout rates; return, length, cost, speed, steering magnitude, action change, and route completion. Environment, reward, action, split, and training metadata are hashed. Comparisons fail closed when task definitions differ.

The primary contrasts are density-0.05 success gain and collision change. Retention is the change in density-0 success and route completion. Density 0.10 measures stress degradation, not a tuning target. Means and sample standard deviations are reported only across completed adaptation lineages; a failed gate remains labeled diagnostic evidence rather than being averaged as zero or hidden.

6 Reproducibility Contract

Each training run writes an immutable resolved configuration, source-lineage record, source checkpoint SHA-256 digest, curriculum state, selected and resumable models, and its own traffic replay buffer. The warm start transfers actor, critic, and target-network parameters into a newly constructed SAC instance; it intentionally does not transfer the source optimizer or traffic-free replay distribution. Observation and action shapes, dtypes, and bounds must match before any parameter is loaded.

The runner may occupy up to four independent MetaDrive subprocesses, but never places multiple en-

gines in one process. A paused traffic run resumes only with its recorded environment count, gradient-step count, reward variant, and stage definition. Completed runs are reused and never overwritten. Atomic meta-data and latest-pointer replacement let a restarted notebook distinguish a resumable run from a mismatched or completed one.

Evaluation artifacts include full episode CSVs and condition summaries. A fingerprint binds map, traffic, horizon, reward, action interface, test split, controller, and training context. Existing results are reused only on an exact fingerprint match. Training also records package and hardware versions plus periodic CPU, RAM, GPU-memory, and throughput samples. Critical artifacts are copied from the live Colab clone to Drive after every long run and verified at the destination.

7 Execution Gates and Public Artifacts

No adaptation begins until two non-learning gates pass. First, native IDMPolicy and ExpertPolicy each run 50 fixed validation scenarios at densities 0.05 and 0.10. This checks task solvability and establishes controller ceilings without treating either hand-engineered policy as a learned baseline. Second, ExpertPolicy produces an actual headless chase-camera frame sequence. A rendering failure stops execution rather than silently substituting an easier top-down view.

The complete workflow lives in one Colab notebook with 20 ordered sections: runtime and Drive setup; compilation and smoke gates; historical summaries; native baselines; frozen source matrices; the two seed-0 pilots; saved variant selection; seed-1 confirmation; final matrices; videos; plots; reports; and final synchronization. Long cells consult latest pointers and curriculum state, so a disconnected runtime can restore artifacts and continue at the next unfinished section.

Five final video roles are declared before seeing their appearance. The frozen and adapted policies share the lowest evaluated density-0.05 scenario seed for a direct before/after comparison. The success video uses the lowest successful adapted scenario; the diagnostic failure uses the lowest failed one when any failure exists. ExpertPolicy provides the native demonstration. Every MP4 has a JSON sidecar with model, selection rule, scenario, density, outcome, return, completion, collision, frame count, camera settings, and experiment fingerprint.

The public artifact set is deliberately small: five comparison figures, seed-level and aggregate CSV/JSON

tables, the selection JSON, selected videos, and this report. This keeps the conclusion tied to the research question instead of turning implementation diagnostics into additional experiments.

8 Predeclared Interpretation

The study separates three claims that are easy to blur. The exposure claim asks whether the adapted policy improves density-0.05 success relative to its own frozen source. The safety-ablation claim asks whether doubling only vehicle and object collision penalties changes the success-collision trade-off under identical seed-0 data and optimization. The retention claim asks whether either improvement preserves the source policy’s traffic-free success and route completion. Density 0.10 is reported as stress generalization and never participates in checkpoint selection.

A low collision rate alone is not positive evidence. A policy that brakes indefinitely will appear collision-free but will lose success, completion, and episode length through timeouts. The qualifying gate therefore requires all four primary behaviors. Conversely, a success increase accompanied by greater collision may be useful learning evidence but does not answer the stated safety question. The report retains both values instead of collapsing them into one reward score.

The reward comparison is exploratory at seed 0 because it selects the confirmation variant. Only the winning rule is confirmed at seed 1, so the final estimate describes two adapted lineages rather than an unbiased two-seed comparison of both rewards. Native controllers establish solvability ceilings, not statistical competitors: they use simulator-specific control logic and do not share SAC’s learned action mapping.

The result remains interpretable if all gates fail. In that case the evidence says that this bounded warm-start and reward intervention did not reliably solve interactive traffic under the allotted 300,000-transition budget. The protocol does not add another seed, tune a third penalty, or replace the architecture after observing failure. This stopping rule is central to keeping a negative result as credible as a positive one.

9 Traffic Results

The traffic-adaptation runs have not yet been executed. No result is imputed from the geometry study or the earlier zero-shot stress test. Once the three declared runs and frozen evaluation matrix finish, the comparison script will generate this section’s macros, tables, and plots directly from the saved episode files.

10 Failure Analysis and Retention

Three outcomes are scientifically useful. First, adaptation may reduce collision while preserving source performance, supporting the proposed transfer. Second, a policy may improve traffic success but lose traffic-free geometry competence, revealing catastrophic forgetting. Third, both reward variants may miss the gate; the predeclared selection still produces one confirmation and reports the miss honestly rather than expanding the search.

The video set uses declared scenario rules. A matched before/after pair uses the same lowest held-out seed. Representative success and diagnostic failure videos use the lowest seed satisfying that outcome in the evaluation CSV. The public default is a moving 1280-by-720 chase view; top-down rendering remains diagnostic only.

11 Limitations

Two training seeds cannot characterize optimizer variance. Traffic density is simulator specific, and built-in traffic policies are not a proxy for human behavior. Procedural maps remain short, observations are privileged low-dimensional simulator state, and validation participates in checkpoint and variant selection. Wilson intervals describe episode sampling uncertainty, not uncertainty across training runs. Most importantly, collision reduction in MetaDrive is not evidence of real-world autonomous-driving safety.

12 Conclusion

SafeRL-Drive first established trustworthy artifacts, then showed that geometry curriculum can materially improve procedural-road generalization. The zero-shot traffic failure provides a concrete final question rather than an excuse for an open-ended experiment sweep. The fixed three-run extension tests warm-start traffic exposure, one controlled reward ablation, retention, and stress degradation. Its value lies in the bounded design and auditable result, whether the final answer is positive or negative.

References

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